Project Design Phase-II
Data Flow Diagram & User Stories

Date	31 January 2025
Team ID	SWTID-2026-7788
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

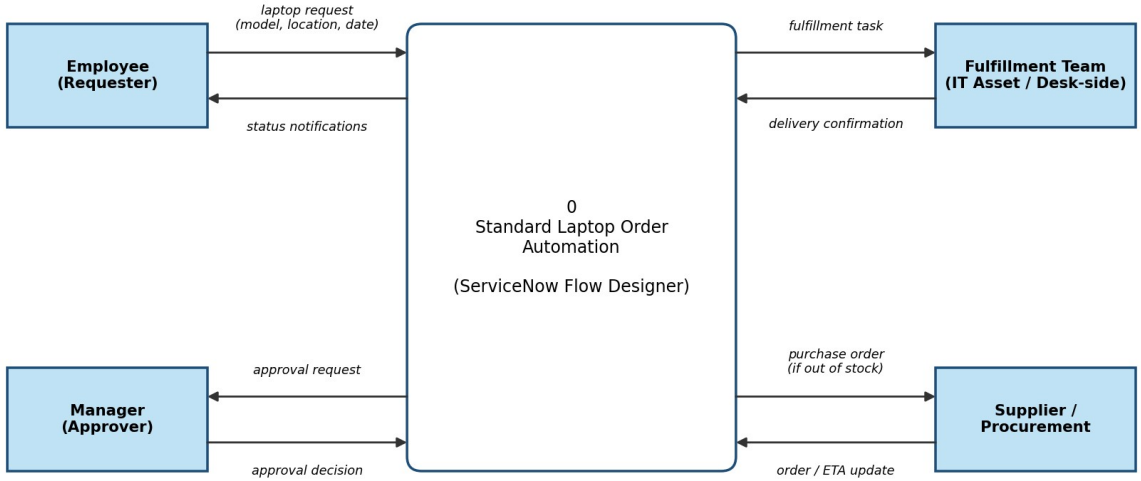
Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

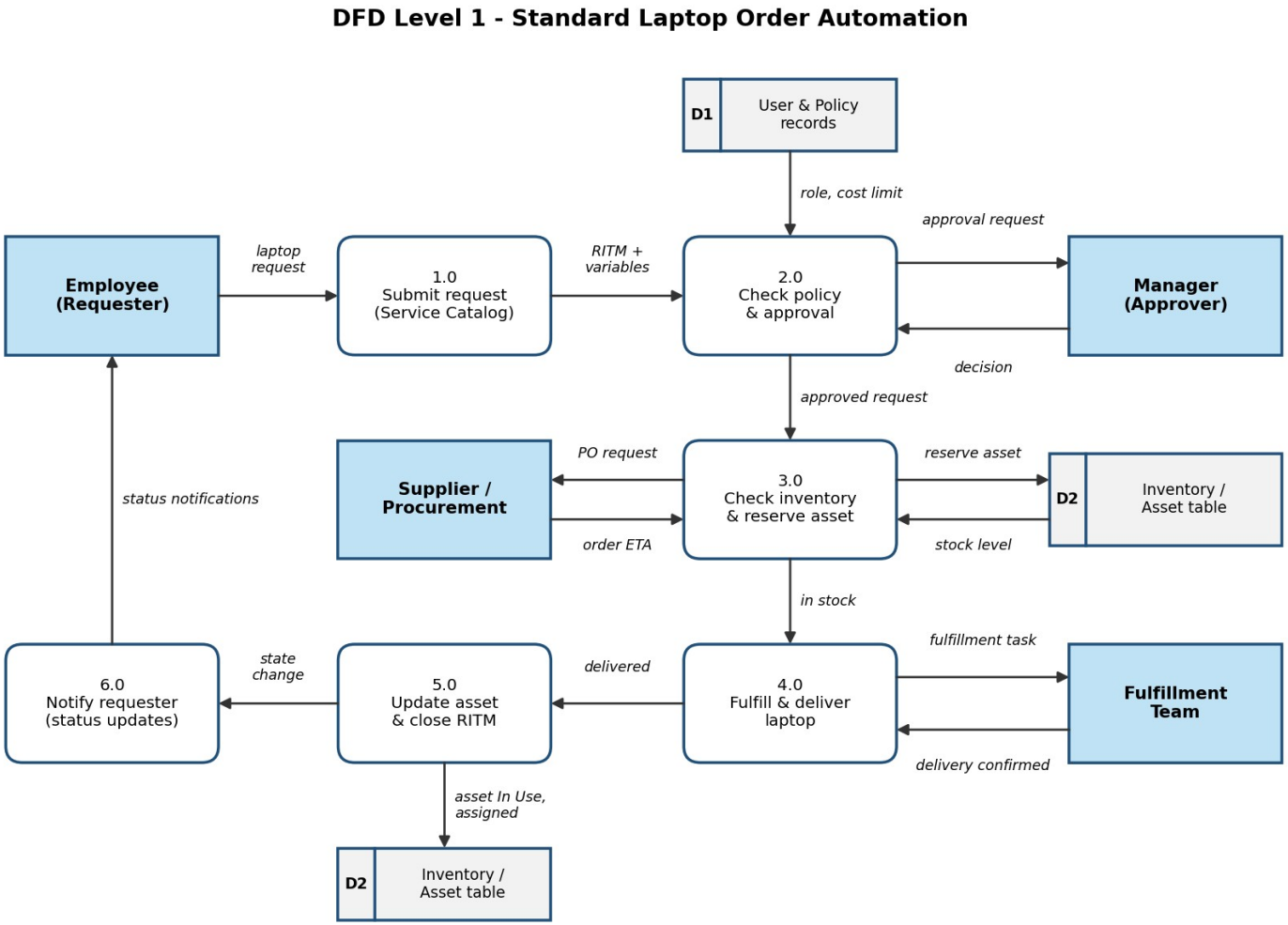
DFD for Standard Laptop Order Automation:

The diagrams below show how a standard laptop request from an employee moves through policy and approval check, inventory check, fulfillment, asset update and status notification in ServiceNow Flow Designer.

DFD Level 0 - Standard Laptop Order Automation



DFD Level 1:


DFD Level 1 - Standard Laptop Order Automation
D1
User & Policy records
role, cost limit
approval request
Employee (Requester)
laptop request
1.0 Submit request (Service Catalog)
RITM + variables
2.0 Check policy & approval
Manager (Approver)
decision
approved request
Supplier / Procurement
PO request
order ETA
3.0 Check inventory & reserve asset
reserve asset
stock level
D2
Inventory / Asset table
status notifications
in stock
fulfillment task
delivered
6.0 Notify requester (status updates)
state change
5.0 Update asset & close RITM
4.0 Fulfill & deliver laptop
Fulfillment Team
delivery confirmed
asset In Use, assigned
D2
Inventory / Asset table

User Stories

User stories for the Standard Laptop Order Automation solution (ServiceNow Flow Designer).

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Employee (Requester)	Request Laptop	USN-1	As an employee, I can find the Standard Laptop item in the Service Catalog so that I can request a laptop.	I can see the Standard Laptop item and open the request form	High	Sprint-1
		USN-2	As an employee, I can see the laptop model pre-selected based on my role and department.	Model is auto-populated from my user record	Medium	Sprint-1
		USN-3	As an employee, I can submit the request by entering only the justification, ship-to location and needed-by date.	Request (RITM) is created and I receive a request number	High	Sprint-1
	Track Order	USN-4	As an employee, I receive a notification at each status change of my request.	I receive an email / notification on approval, shipment and delivery	High	Sprint-2
		USN-5	As an employee, I can view the current status of my request in the portal.	Status is visible under My Requests	Medium	Sprint-2
	Feedback	USN-6	As an employee, I receive a satisfaction survey after my request is closed.	Survey is sent automatically when the RITM is closed	Low	Sprint-3
Manager (Approver)	Approval	USN-7	As a manager, I receive an approval request only when the laptop cost is above the policy threshold.	Requests within policy are auto-approved; others go to my approval queue	High	Sprint-1
		USN-8	As a manager, I can approve or reject a request from email or the portal and add comments.	Decision is recorded on the RITM and the requester is notified	High	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
IT Fulfillment Team	Fulfillment	USN-9	As a fulfillment agent, I get a fulfillment task automatically when stock is available.	Task is created for my group with the asset already reserved	High	Sprint-2
		USN-10	As a fulfillment agent, I can mark the laptop as delivered and the asset is assigned to the employee automatically.	Asset state is In Use, assigned to the user, and the RITM is closed	High	Sprint-2
IT Asset Manager / Procurement	Inventory	USN-11	As an asset manager, I want the flow to check stock automatically for every approved request.	Stock is looked up on the asset table and reserved when available	High	Sprint-1
		USN-12	As an asset manager, I want a purchase requisition to be created automatically when stock is not available.	PO is linked to the RITM and the requester is told about the delay	Medium	Sprint-3
Administrator	Monitoring & Reporting	USN-13	As an administrator, I want rejected requests and flow failures routed to an IT queue.	A task is created in the IT queue for every rejection or flow error	Medium	Sprint-3
		USN-14	As an administrator, I can view a dashboard of request cycle time and stock trends.	Dashboard shows average cycle time and current stock levels	Low	Sprint-3